

Dear students here is the list of theorems for the final. You should study their proof! Remember it cannot be included in your cheatsheet.

1. Characterization of cyclic groups: Let $(G, \cdot, e_G)$ be a cyclic group, then:
 - If G is infinite then $G \cong (\mathbb{Z}, +, 0)$.
 - If G is finite then $G \cong (\mathbb{Z}/k\mathbb{Z}, +_k, [0]_k)$ where k is the order of the generator of G .
2. Characterization of subgroups of cyclic groups: Let $(G, \cdot, e_G)$ be a cyclic group, then:
 - If G is infinite, then any subgroup $H \leq G$ is of the form $(G^m, \cdot, e_G)$ for some $m \in \mathbb{N}$.
 - If G is finite of order n . Assume that $G = \langle a \rangle$. Then any subgroup H of G is of the form $\langle a^s \rangle$ where $s < n$. Moreover, $\langle a^s \rangle = \langle a^t \rangle$ if and only if $HCD(t, n) = HCD(s, n)$.
3. Cayley's theorem: Let $(G, \cdot, e_G)$ then G is isomorphic to a subgroup of $(\Sigma(G), \circ, id_G)$ (the group of permutation of G).
4. Decomposition of permutation in disjoint cycles: Let S_n be the group of permutations. Given $\sigma \in S_n$ there are t_i finitely many disjoint cycles such that $\sigma = \prod_{i=1}^k \tau_i$.
5. Invariance of parity of permutations: Given $\sigma \in S_n$ then σ can be written as a product of an odd number of transpositions, or an even number of transpositions but not both.
6. Correspondance between normal subgroups and kernel of homomorphism: Let $(G, \cdot, e_G)$ be a group.

Then if N is a normal subgroup of G , there is a group $(K, \cdot_K, e_K)$ and an homomorphism $\phi : G \mapsto K$ such that $N = Ker(\phi)$.

 - Let $(K, \cdot_K, e_K)$ be a group and $\phi : G \mapsto K$ be an homomorphism. Then $Ker(\phi)$ is a normal subgroup of G .
7. Isomorphism theorems:
 - (a) *First isomorphism theorem*: Let $(G, \cdot_G, e_G)$ and $(K, \cdot_K, e_K)$ be groups and $\phi : G \rightarrow K$ be a surjective homomorphism. Then $G/Ker(\phi) \cong K$.
 - (b) *Second Isomorphism theorem*: Let $(G, \cdot_G, e_G)$ be a group and $H \leq G$ be a subgroup and $N \leq G$ be a normal subgroup. Then:
 - HN is a subgroup of G ,
 - $HN/N \cong H/(H \cap N)$.
 - (c) *Third Isomorphism theorem*: Let $(G, \cdot_G, e_G)$ be a group. Let H and N be normal subgroups of G with $K \leq H$. Then $G/H \cong (G/K)/(H/K)$.

8. Sylow theorems: Let G be a finite group and p a prime. Assume that $|G| = p^n m$ where p does not divide m .
- (a) *First theorem:*
- For each $1 \leq i \leq n$ there is a subgroup H_i of G with p^i -elements.
 - Every subgroup H_i of G of p^i -elements is a normal subgroup of a group H_{i+1} with p^{i+1} -elements.
9. *Second theorem:* Let P_1 and P_2 p -syllow groups of G , then there are conjugate i.e. there is $g \in G$ such that $P_1 = gP_2g^{-1}$.
10. *Third Isomorphism theorem:* Let n_p be the number of p -syllow groups of G . Then n_p divides $|G|$ and $n_p \equiv_p 1$.